

Paper #75 — Honest Assessment (R-13)

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May 5, 2026

Paper #75: RH via Selberg Class — Honest Assessment

Applying the same T1465 discipline to Paper #75 that we applied to Paper #88 for BSD.

What IS Right (genuinely publishable structure)

1. The reduction strategy is real: Reducing RH to a finite spectral condition on $\Gamma(137)\backslash\mathrm{SO}_0(5,2)$ is a legitimate mathematical program. The framework (automorphic spectral geometry + Arthur packets + temperedness forcing) is correctly composed from established results.
 2. The numerical evidence is strong: 57/57 tests PASS. Epstein negative test: 9/9 correct rejections. The spectral structure computationally works.
 3. The Selberg class embedding is sound: For degree $d_F \leq 2$, embedding into $\mathrm{SO}(5,2)$ representations is standard (Kim-Shahidi, Cogdell-Piatetski-Shapiro). The approach has precedent.
 4. The Arthur packet taxonomy is correct: 45 non-tempered types enumerated for $\mathrm{SO}(7)$ follows Arthur's classification. The seven constraints correspond to genuine spectral data.
 5. The paper is well-written and would receive serious attention from experts.
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Three Critical Gaps (named, with tiers)

Gap A: The 91.1 Spectral Gap Citation (CRITICAL — Tier C)

Claim: $\lambda_1(\Gamma(137)\backslash\mathrm{SO}_0(5,2)) \geq 91.1$, citing [PS09] (Pitale-Schmidt 2009).

Problem: [PS09] proves bounds for p-adic representations of $\mathrm{GSp}(4)$, not global eigenvalue bounds on $\mathrm{SO}_0(5,2)$. $\mathrm{GSp}(4)$ is the split form; $\mathrm{SO}_0(5,2)$ is the real form relevant to our locally symmetric space. These are related by functorial transfer, but the bound does NOT transfer directly.

What would fix it: - Find a spectral gap bound specifically for $\Gamma(N)\backslash\mathrm{SO}_0(5,2)$ (Bergeron-Clozel style)
- OR: prove the transfer from $\mathrm{GSp}(4)$ to $\mathrm{SO}(5,2)$ preserves the bound - OR: compute λ_1 directly for $\Gamma(137)\backslash D_{IV}^5$ (doable in principle — finite computation on a definite space)

Tier: C (conditional on unproved spectral gap transfer)

Owner: R-9 (Lyra, URGENT)

Gap B: L-function Degree Mismatch (CRITICAL — Tier C)

Claim: Theorem 6.1 Step 1 asserts $L(s, \pi_F, \mathrm{std}) = F(s)$ where π_F is an automorphic representation of $\mathrm{SO}(7)$.

Problem: The standard L-function of $SO(7)$ has degree 7 (the dimension of the standard representation of the Langlands dual $Sp(6)$). But $\zeta(s)$ has degree 1. How does degree 1 factor out of degree 7?

What would fix it: - Explicit Rankin-Selberg decomposition showing which factor of the degree-7 L-function recovers $\zeta(s)$ - OR: use a DIFFERENT representation (not standard) — e.g., the spin representation (degree 8) decomposes differently - OR: clarify that we work with a SPECIFIC functorial lift from $GL(1)$ to $SO(7)$ where the embedding naturally produces the degree-1 factor

Tier: C (conditional on explicit degree decomposition)

Owner: R-10 (Lyra+Elie, URGENT)

Gap C: Constraint 1 Parity Computation (SERIOUS — Tier C)

Claim: $\chi_{\pi}(-1) = (-1)^{p+q}$ for $SO(p,q)$ representations. Used to kill 34/45 Arthur packets.

Problem: This sign formula is not in Arthur's book "The Endoscopic Classification of Representations" (2013). Standard parity for tempered packets involves root numbers and L-packets, not a simple $(-1)^{p+q}$ formula. The claim needs either: - A citation to the specific result in representation theory - A proof as a lemma (from Weyl character formula or Schur indicator) - OR: removal with audit of which packets remain unkilld

Tier: C (conditional on unproved sign computation)

Owner: R-11 (Lyra, URGENT)

Honest Status

Component	Status	Tier
Strategy (reduce RH to spectral condition)	Sound	D
Arthur packet taxonomy (45 types)	Correct	D
Numerical verification (57/57)	Strong	D
Spectral gap ≥ 91.1	WRONG for $SO(5,2)$ but UNNECESSARY if R-11 done	C ☐ moot
L-function recovery (degree 6 factorization)	RESOLVED by Elie (F is factor of Std_6)	C ☐ D
Constraint 1 (parity kills 34/45)	UNPROVED — essential if Constraint 2 removed	C
Thm 6.1 Step 3 (temperedness ☐ GRH)	NEW GAP — conflates spectral params with L-function zeros	C

Overall (revised by Lyra, May 5): The proof has TWO independent conditionals: 1. Temperedness: All representations on $\Gamma(137) \backslash D_{IV}^5$ are tempered. Conditional on R-11 (parity sign). R-9 is unnecessary if R-11 resolved. 2. Step 3: Temperedness implies GRH for the standard L-function. This is NOT a theorem — it conflates spectral parameters (which temperedness constrains) with zeros of $L(s, \pi, \text{Std}_6)$ in the critical strip (which are properties of the analytic continuation). Fix requires either Selberg trace formula argument or is itself conditional.

The paper unconditionally proves: Arthur packet elimination (Sections 3-4, pending R-11). The paper conditionally proves: RH, conditional on (R-11) + (temperedness ☐ GRH).

Recommended Path (revised by Lyra)

1. DO NOT submit to Annals until R-10 Step 3 and R-11 resolved.
2. R-11 is tractable: Finite computation of Arthur signs for $SO(5,2)$. Do this first.
3. R-9 is moot: Once R-11 done, Constraints $\{1,3\}$ eliminate all 45 types. No need for 91.1.
4. R-10 Step 3 is the binding constraint: Options:
 - (a) Rewrite Section 6 using Selberg trace formula (relates zeros of zeta directly to spectral parameters)
 - (b) Prove “temperedness $\Rightarrow$ GRH” for degree-6 L-functions of tempered $SO(7)$ reps (open problem)
 - (c) Reframe: “Temperedness of $\Gamma(137)\backslash D_{IV}^5$ is unconditional; RH follows IF temperedness $\Rightarrow$ GRH” (honest, publishable)
5. Circulate to Sarnak/Gan/Ichino with SPECIFIC questions about (a) Selberg trace formula for type IV domains and (b) whether temperedness-implies-GRH is known for any cases.

Comparison to BSD Assessment (Paper #88)

	Paper #88 (BSD)	Paper #75 (RH)
Core mechanism	Chern hole (real math)	Arthur packets (real math)
Gap type	Labeling (DOF-to-K-type transfer)	Step 3: temperedness $\neq$ GRH
Fixability	HIGH (one standalone lemma)	UNCERTAIN (may need Selberg trace formula rewrite)
Numerical evidence	T1426 (51 curves, 0 exceptions)	57/57 tests PASS
Honest label	99.7% (conditional on dictionary)	CONDITIONAL (R-11 + Step 3)
Action	Submit after R-2 accepted	Do NOT submit until R-10 Step 3 resolved

BSD’s gap is a labeling issue — the math works, we just need to prove one transfer lemma. RH’s gap is now understood to have two layers: the elimination machinery (fixable via R-11) and the final inference (temperedness $\Rightarrow$ GRH, which may require a fundamentally different argument from what the paper presents).

Written by Keeper, May 5, 2026. Updated by Lyra, May 5, 2026 (~10 AM) with deeper R-10 Step 3 analysis. See [notes/BST_Paper75_R9_R10_R11_Analysis.md](#) for full technical details.